Lesson 9.2 - How to Find the Z-score Given the Confidence Intervals

> **Video:** How To Find The Z Score Given The Confidence Level of a Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Common values to memorize

Confidence level	z
90%	1.645
95%	1.96
98%	2.33
99%	2.575

Method for any confidence level

Step 1 - area to the left of $+z$:

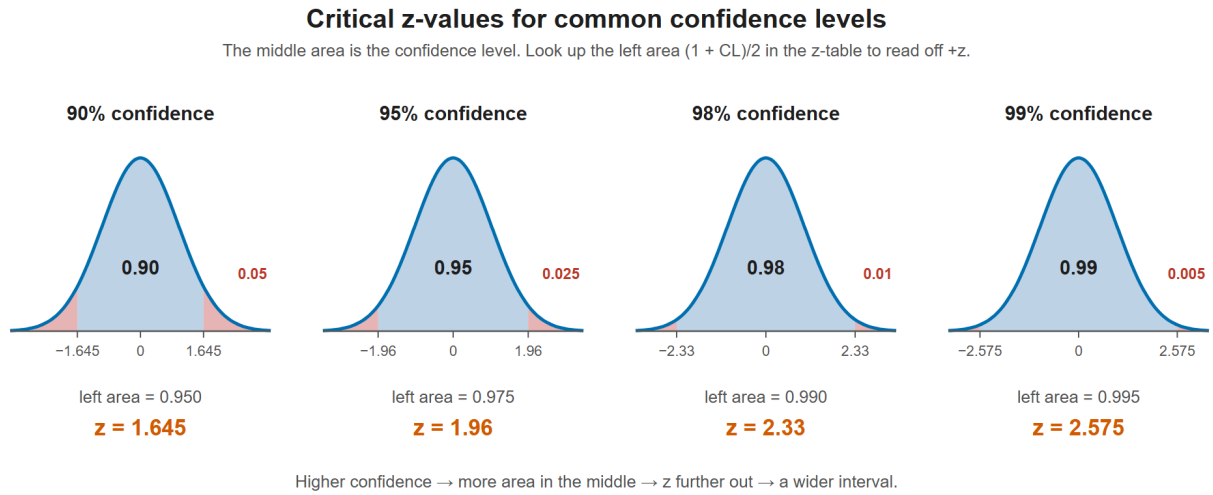
$$\text{left area} = \frac{1 + \text{CL}}{2}$$

Why: the confidence level is the area between $-z$ and $+z$. The leftover $1 - \text{CL}$ is split evenly between the two tails. For 95%:

```
 0.025  |          0.95           |  0.025
--------|-------------------------|--------
       -z           mean         +z
left area up to +z = 0.95 + 0.025 = 0.975
```

Step 2 - find that area inside a positive z-table (one that gives **left** areas). Read the row (ones and tenths digits) and the column (hundredths digit), then add them.

Example: 0.975 sits in row **1.9**, column **0.06**, so $z = 1.9 + 0.06 = 1.96$.



The confidence level is the middle area, and each tail gets half of what is left. The left area up to +z is (1 + CL)/2, which is the number to look up in the z-table.

Practice

97% confidence

$$\text{left area} = \frac{1 + \text{CL}}{2} = \frac{1 + 0.97}{2} = 0.985 \;\Rightarrow\; \text{row } 2.1,\ \text{column } 0.07 \;\Rightarrow\; z = 2.17$$

92% confidence

$$\text{left area} = \frac{1 + \text{CL}}{2} = \frac{1 + 0.92}{2} = 0.96$$

The closest table entry is 0.95994, in row **1.7**, column **0.05**, so $z = 1.75$.

Key takeaways

- Left area $= (1 + \text{CL})/2$. Look it up in the body of the z-table to read off z.
- If the exact area isn't in the table, use the closest entry, or average two entries that are equally close (e.g. 90% → 1.645).
- Know 1.645, 1.96, 2.33 and 2.575 by heart.